

Sequential Covering Pseudocode

Inputs: labeled training dataset D

Outputs: ruleset R that covers all instances in D

Procedure:

Initialize R as the empty set

for each class C {

 while D is nonempty {

 Construct one rule r that correctly classifies
 some instances in D that belong to class C and
 does not incorrectly classify any non-C instances

 Add rule r to ruleset R

 Remove from D all instances correctly classified by r

 }

}

return R

Finding the next rule

Notice that the sequential covering pseudocode does not specify how the rule r is to be constructed on a given pass through the inner while loop. Different sequential covering algorithms do this in different ways. The PRISM algorithm uses a depth-first search to construct the next rule for a given class C . Since the consequent of the rule must be the given class C , only the antecedent needs to be constructed; this is done by starting with an empty antecedent and iteratively adding the most promising attribute=value constraint next. The classification accuracy is used to rate candidate rules. This depth-first search continues until the resulting rule is specific enough that it makes no classification errors over the available data instances. Pseudocode follows below.

Inputs: nonempty dataset D , class C

Outputs: a single rule r with C as its consequent that covers some instances in D and that does not incorrectly classify any non- C instances in D

Procedure:

Initialize r as a rule with an empty antecedent, which predicts C without examining any attributes:

$\Rightarrow C$

Initialize A as the set of all attributes over D

while r incorrectly classifies some non- C instances of D {
 write r as $\text{ant}(r) \Rightarrow C$

 for each attribute-value pair $(a=v)$, where a belongs to A and v is a value of a , compute the accuracy of the rule $\text{ant}(r)$ and $(a=v) \Rightarrow C$

 let $(a^*=v^*)$ be the attribute-value pair of maximum accuracy over D ; in case of a tie, choose the pair that covers the greatest number of instances of D

 update r by adding $(a^*=v^*)$ to its antecedent:
 $r = (\text{ant}(r) \text{ and } (a^*=v^*)) \Rightarrow C$

 remove the attribute a^* from the set A :
 $A = A - \{a^*\}$

 }
return r

% 9. Class Distribution:

- % 1. hard contact lenses: 4
- % 2. soft contact lenses: 5
- % 3. no contact lenses: 15

@relation contact-lenses

@attribute age {young, pre-presbyopic, presbyopic}
@attribute spectacle-prescrip {myope, hypermetrope}
@attribute astigmatism {no, yes}
@attribute tear-prod-rate {reduced, normal}
@attribute contact-lenses {soft, hard, none}

@data

%
% 24 instances
%

Age	spectacle-prescrip	astigmatism	tear-prod-rate	contact-lenses
Young	myope	No	reduced	none
Young	myope	No	normal	soft
Young	myope	Yes	reduced	none
Young	myope	Yes	normal	hard
Young	hypermetrope	No	reduced	none
Young	hypermetrope	No	normal	soft
Young	hypermetrope	Yes	reduced	none
Young	hypermetrope	Yes	normal	hard
pre-presbyopic	myope	No	reduced	none
pre-presbyopic	myope	No	normal	soft
pre-presbyopic	myope	Yes	reduced	none
pre-presbyopic	myope	Yes	normal	hard
pre-presbyopic	hypermetrope	No	reduced	none
pre-presbyopic	hypermetrope	No	normal	soft
pre-presbyopic	hypermetrope	Yes	reduced	none
pre-presbyopic	hypermetrope	Yes	normal	none
presbyopic	myope	No	reduced	none
presbyopic	myope	No	normal	none
presbyopic	myope	Yes	reduced	none
presbyopic	myope	Yes	normal	hard
presbyopic	hypermetrope	No	reduced	none
presbyopic	hypermetrope	No	normal	soft
presbyopic	hypermetrope	Yes	reduced	none
presbyopic	hypermetrope	Yes	normal	none

Finding the first rule

The main loop in the sequential covering pseudocode first chooses contact-lenses=hard as the class. Work then focuses on finding one rule for the hard class.

Initialization

Initially, the rule is just

$\Rightarrow \text{hard}$

and all non-class attributes are still being considered, that is,

$A = \{ \text{age, spectacle-prescription, astigmatism, tear-production-rate} \}$

First pass through the loop

The accuracies are computed for all candidate rules of the form:

$(a = v) \Rightarrow C$

where (a, v) ranges over all possible attribute-value pairs:

attribute-value pair	p/t
age=young	2/8
age=pre-presbyopic	1/8
age=presbyopic	1/8
spectacle-prescription=myope	3/12
spectacle-prescription=hypermetrope	1/12
astigmatism=no	0/12
astigmatism=yes	4/12
tear-production-rate=normal	4/12
tear-production-rate=reduced	0/12

In each case, the denominator of the fraction p/t on the right indicates the number of instances that contain the given attribute-value pair, and the numerator is the number of these instances that belong to the hard class. For example, there are 8 instances with age=young, and 2 of these 8 are in the hard class. The attribute-value pairs with the largest accuracies are astigmatism=yes and tear-production-rate=normal. Since both of these have the same coverage, one of them is chosen arbitrarily, say astigmatism=yes. That is:

$(a^*, v^*) = (\text{astigmatism}, \text{yes})$

The rule r is then updated by adding this attribute-value pair to the antecedent:

$r = (\text{astigmatism}=\text{yes}) \Rightarrow \text{hard}$

and the set of attributes being considered is reduced by removing the astigmatism attribute:

$A = \{ \text{age, spectacle-prescription, tear-production-rate} \}$

Since the current rule r incorrectly classifies some instances by predicting hard for some instances that are actually in the contact-lenses=none class, another pass is made through the loop that refines the rule.

Second pass

Candidate rules for this pass are of the form

(astigmatism=yes) and (a=v) => hard

Accuracies for the candidate attribute-value pairs are:

attribute-value pair	p/t
age=young	2/4
age=pre-presbyopic	1/4
age=presbyopic	1/4
spectacle-prescription=myope	3/6
spectacle-prescription=hypermetrope	1/6
tear-production-rate=normal	4/6
tear-production-rate=reduced	0/6

In computing these accuracies, only instances for which (astigmatism=yes) are considered.

Here is the set of all such instances:

young	myope	yes	reduced	none
young	myope	yes	normal	hard
young	hypermetrope	yes	reduced	none
young	hypermetrope	yes	normal	hard
pre-presbyopic	myope	yes	reduced	none
pre-presbyopic	myope	yes	normal	hard
pre-presbyopic	hypermetrope	yes	reduced	none
pre-presbyopic	hypermetrope	yes	normal	none
presbyopic	myope	yes	reduced	none
presbyopic	myope	yes	normal	hard
presbyopic	hypermetrope	yes	reduced	none
presbyopic	hypermetrope	yes	normal	none

The attribute-value pair of greatest accuracy is tear-production-rate=normal, with an accuracy p/t = 4/6. The rule r is updated by adding this attribute-value pair to the antecedent:

r = (astigmatism=yes) and (tear-production-rate=normal) => hard

A = { age, spectacle-prescription } Some instances are still misclassified by the refined rule r, so another pass is made through the loop.

Third pass

Candidate rules are of the form:

r = (astigmatism=yes) and (tear-production-rate=normal) and (a=v) => hard

Accuracies for the candidate attribute-value pairs are:

attribute-value pair	p/t
age=young	2/2
age=pre-presbyopic	1/2
age=presbyopic	1/2
spectacle-prescription=myope	3/3
spectacle-prescription=hypermetrope	1/3

These are computed over the set of instances that satisfy both constraints that appear in the current antecedent:

young	myope	yes	normal	hard
young	hypermetrope	yes	normal	hard
pre-presbyopic	myope	yes	normal	hard
pre-presbyopic	hypermetrope	yes	normal	none
presbyopic	myope	yes	normal	hard
presbyopic	hypermetrope	yes	normal	none

There are now two attribute-value pairs with 100% accuracy, namely (age=young) and (spectacle-prescription=myope). The tie is broken by choosing the one with greater coverage, that is:

$(a^*, v^*) = (\text{spectacle-prescription}, \text{myope})$

The updated rule is

$r = (\text{astigmatism}=\text{yes}) \text{ and } (\text{tear-production-rate}=\text{normal}) \text{ and } (\text{spectacle-prescription}=\text{myope}) \Rightarrow \text{hard}$

Since this rule has 100% accuracy, it is accepted and returned by the rule construction procedure as the first rule generated for the hard class by the inner loop in the sequential covering pseudocode.

Updates in the sequential covering inner loop

The first rule

$r = (\text{astigmatism}=\text{yes}) \text{ and } (\text{tear-production-rate}=\text{normal}) \text{ and } (\text{spectacle-prescription}=\text{myope}) \Rightarrow \text{hard}$

is added to the ruleset R:

$R = \{ (\text{astigmatism}=\text{yes}) \text{ and } (\text{tear-production-rate}=\text{normal}) \text{ and } (\text{spectacle-prescription}=\text{myope}) \Rightarrow \text{hard} \}$

and the three instances covered by this rule are removed from the dataset D. D now contains 21 instances:

young	Myope	no	reduced	none
young	Myope	no	normal	soft
young	Myope	yes	reduced	none
young	hypermetrope	no	reduced	none
young	hypermetrope	no	normal	soft
young	hypermetrope	yes	reduced	none
young	hypermetrope	yes	normal	hard
pre-presbyopic	Myope	no	reduced	none
pre-presbyopic	Myope	no	normal	soft
pre-presbyopic	Myope	yes	reduced	none
pre-presbyopic	hypermetrope	no	reduced	none
pre-presbyopic	hypermetrope	no	normal	soft
pre-presbyopic	hypermetrope	yes	reduced	none
pre-presbyopic	hypermetrope	yes	normal	none
presbyopic	Myope	no	reduced	none
presbyopic	Myope	no	normal	none
presbyopic	Myope	yes	reduced	none
presbyopic	hypermetrope	no	reduced	none
presbyopic	hypermetrope	no	normal	soft
presbyopic	hypermetrope	yes	reduced	none
presbyopic	hypermetrope	yes	normal	none

This completes the first pass through the inner loop in the sequential covering algorithm. Since one instance with contact-lenses=hard still remains, another pass is made in order to generate a second rule that targets the hard class. This second rule is constructed by following the procedure described above. The second rule turns out to be:

$r = (\text{age}=\text{young}) \text{ and } (\text{astigmatism}=\text{yes}) \text{ and } (\text{tear-production-rate}=\text{normal}) \Rightarrow \text{hard}$

This rule covers the remaining instance in the hard class, which completes the first pass through the outer loop of the sequential covering algorithm. Subsequent passes generate rules that cover instances belonging to the soft and none classes.